

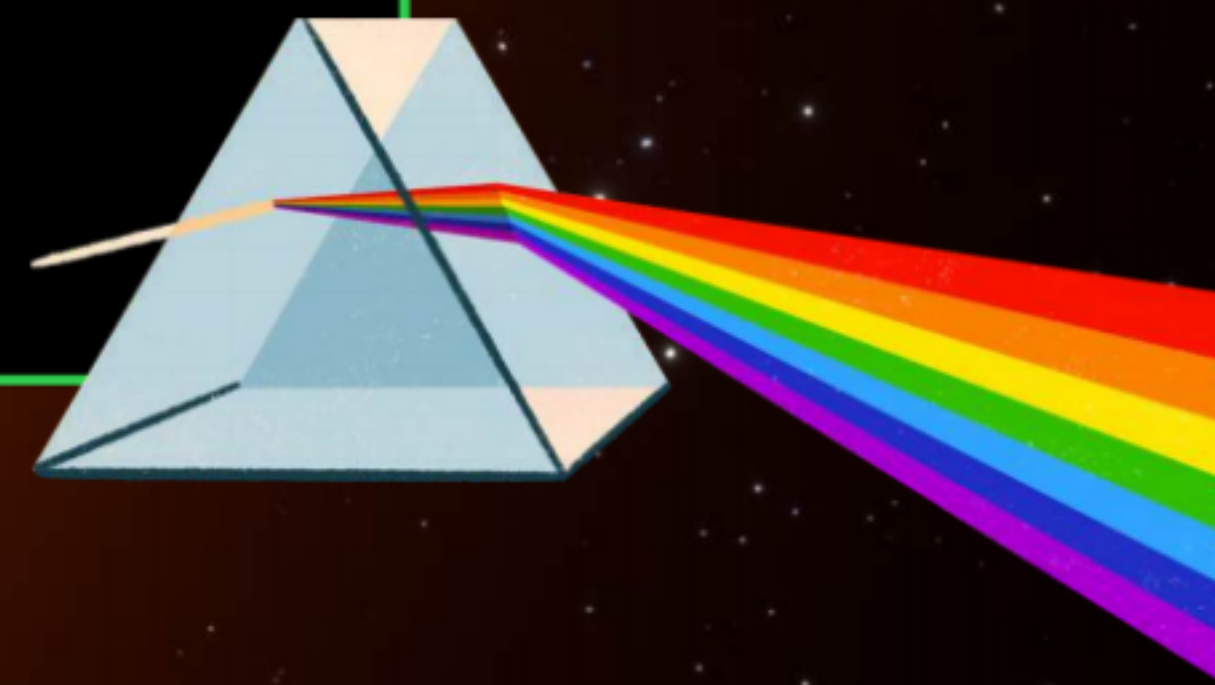
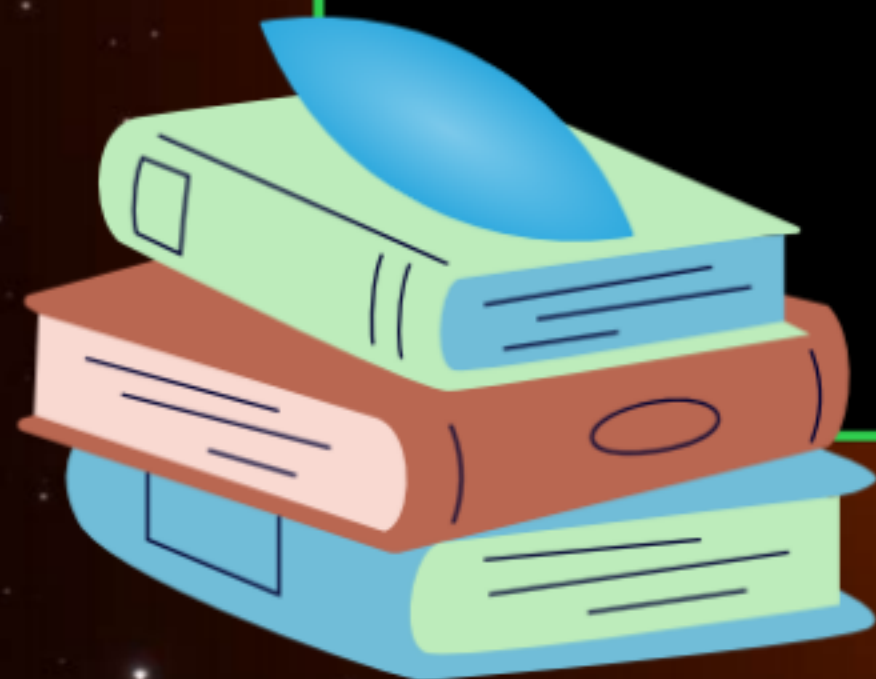
L4

आरंभ
BATCH

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD



Today's Targets!



- Numericals



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$m = \ominus \quad \oplus$$

Real Virtual



SOCHO!

Q. An optical device has been given to a student and he determines its focal length by focusing the image of the sun on a screen placed 24 cm from the device on the same side as the sun. Select the correct statement about the device.

- (a) Convex mirror of focal length 12 cm.
- (b) Convex lens of focal length 24 cm.
- ☒ (c) Concave mirror of focal length 24 cm.
- (d) Convex lens of focal length 12 cm.





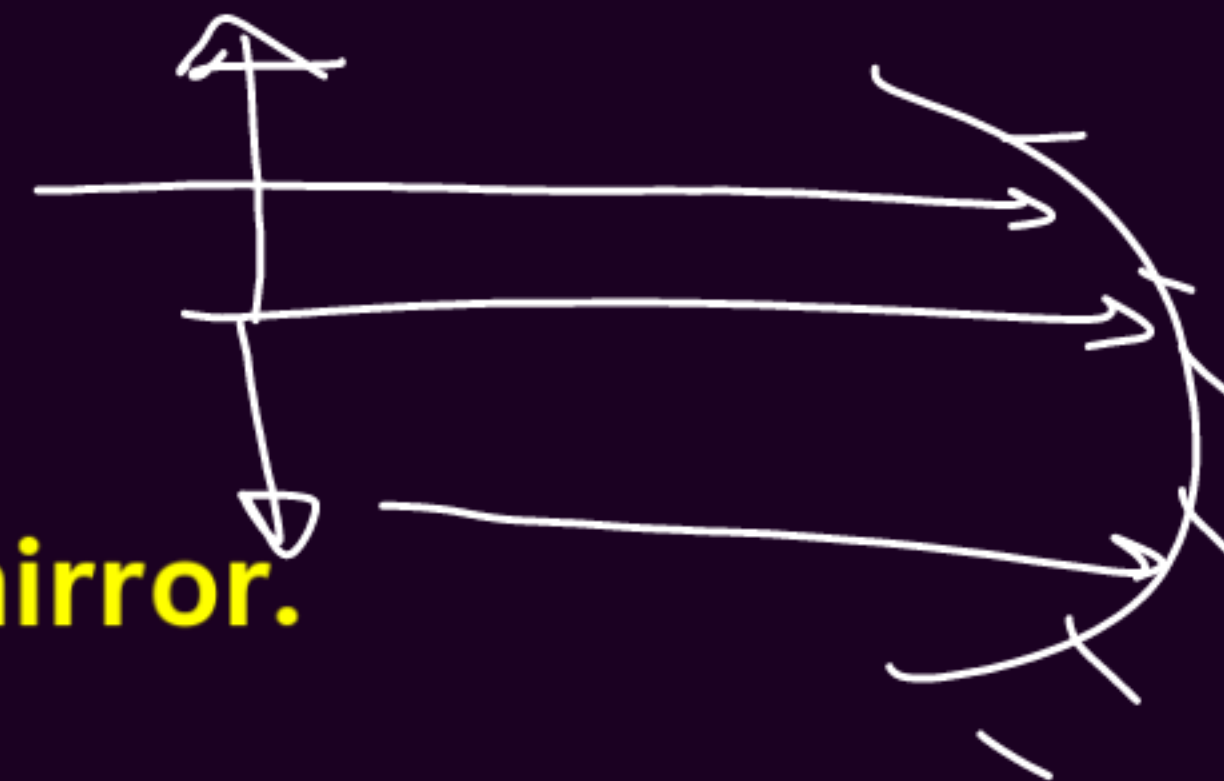
SOCHO!

$C \rightarrow 30\text{cm}$

focal = 15 cm

Q. Rays from the sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that the size of its image is equal to the size of the object?

- (a) 15 cm in front of the mirror.
- ✓ (b) 30 cm in front of the mirror.
- (c) Between 15 cm and 30 cm in front of the mirror.
- (d) More than 30 cm in front of the mirror.





SOCHO!

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{u} + \left[\frac{-3}{u} \right] = \frac{1}{20}$$

$$\frac{-2}{u} = \frac{1}{20}$$

$$\Rightarrow u = -40$$

Q. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is +1/2. Where should the object be placed to reduce the magnification to +1/3?

$$m' = -\frac{v}{u}$$

$$\frac{1}{3} = -\frac{v}{u}$$

$$\frac{u}{3} = -v$$

$$v = -\frac{u}{3}$$

$$m = +1/2$$

$$u = -20 \text{ cm}$$

$$m' = +1/3$$

$$m = -\frac{v}{u}$$

$$\frac{1}{2} = \frac{-v}{-20}$$

$$v = +10 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{-1}{20} + \frac{1}{10}$$

$$= \frac{-1+2}{20} = \frac{1}{20}$$

$$u = 20 \text{ cm}$$



SOCHO!



Q. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use the mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer.

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{u} - \frac{1}{24} = -\frac{1}{18}$$

$$\frac{2-1}{24} = -\frac{1}{18}$$

$$f = -18 \text{ cm} \Rightarrow 0-18 \text{ cm} \text{ Less than } 18 \text{ cm}$$

$$m = +2 = -\frac{v}{u}$$

$$v = -2u$$

$$\frac{18}{-90} = \frac{1}{u}$$



SOCHO!



Q. The image of a candle flame placed at a distance of 30 cm from a mirror is formed on a screen placed in front of the mirror at a distance of 60 cm from its pole. What is the nature of the mirror? Find its focal length. If the height of the flame is 2.4 cm, find the height of its image. State whether the image formed is erect or inverted.

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$= -\frac{1}{30} - \frac{1}{60} = -\frac{2}{60}$$

$$f = -20\text{cm}$$

$$u = -30\text{cm}$$

$$v = -60\text{cm}$$

→ Concave

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = -\left(\frac{-60}{-30}\right) \Rightarrow m = -2$$

$$h_o = 2.4\text{cm}$$

$$h_i = ?$$

$$m = -2 = \frac{h_i}{h_o} = \frac{h_i}{2.4}$$

$$h_i = -4.8\text{cm}$$



SOCHO!

-30cm, 8cm

Q. An object 4 cm in height, is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed to obtain a sharp image of the object. Calculate the height of the image

$$\begin{aligned}h_o &= 4\text{cm} \\u &= -15\text{cm} \\f &= -10\text{cm} \\v &= ?\end{aligned}$$

$$h_i = ?$$

$$m = -\frac{v}{u} = \frac{h_i}{h_o}$$



SOCHO!



Q. Rohit wants to have an erect image of an object using a converging mirror of focal length 40 cm. *→ Virtual*

(A) Help Rohit by specifying the range of distance where the object can be placed in front of the mirror. Justify by providing a reason for the same. *→ 0-40cm*

(B) Draw a ray diagram to show image formation in this case.

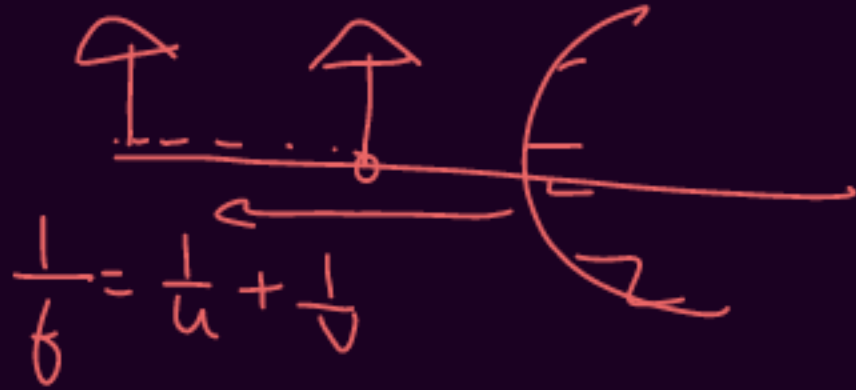
(C) State one use of the mirror based on the above kind of image formation.

→ Optical Instrument



SOCHO!

Q. Convex mirrors are commonly used as passenger-side mirrors in vehicles and at places like road crossings, hallways, warehouses, and sharp turns. They provide a wider field of view, reduce blind spots, and improve safety by allowing people to see approaching vehicles or individuals.



$$m = -\frac{v}{u}$$

$$m = 0.2$$

In workplaces and production units, convex mirrors help monitor movement and inspect products on conveyor belts, thereby increasing safety and efficiency.

(A) As the object is moved away from a convex mirror, what will be the effect on the size of the image?

(B) When a car moves away from a convex mirror installed at a road crossing, what happens to the size of its image?

(C) Consider an object 7.0 cm in length which is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. What is the size and position of image formed?

$$h_o = 7 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$R = 30 \text{ cm}$$

$$f = +15 \text{ cm}$$

$$v = ?$$

$$h_i = ?$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{15} = -\frac{1}{20} + \frac{1}{v}$$

$$\frac{1}{15} + \frac{1}{20} = \frac{1}{v}$$

$$\frac{4+3}{60} = \frac{1}{v}$$

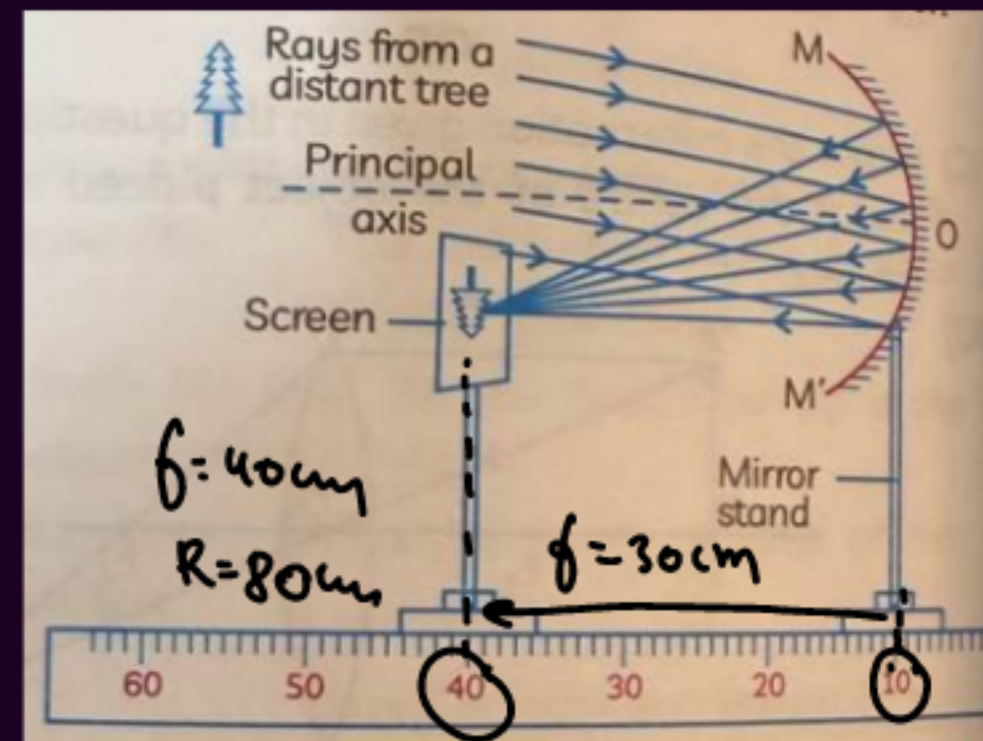
$$v = \frac{60}{7} \text{ cm}$$



SOCHO!

Q. Concave mirrors are used in many ways in daily life. They are used in torches, headlights and searchlights to produce a strong parallel beam of light. They are also used as shaving mirrors to form an erect and enlarged image of the face.

A student had a concave mirror and wanted to find its focal length. So, he performed an activity by focusing rays from a distant object on a screen.



(A) The radius of curvature of the concave mirror used by the student in the activity?

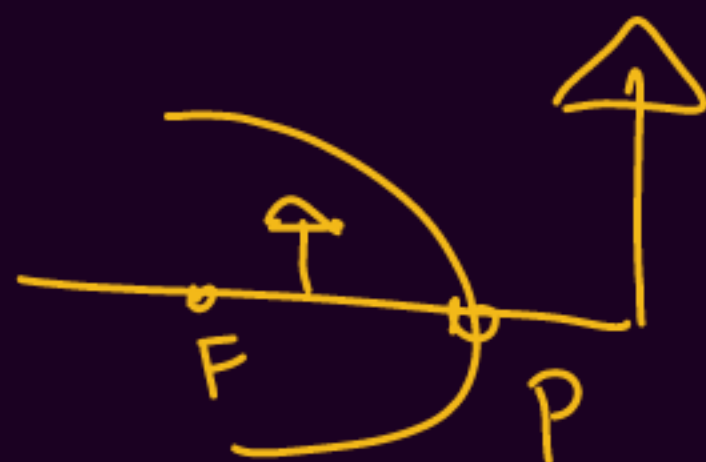
- (a) 60 cm
- (b) 40 cm
- (c) 30 cm
- (d) 10 cm



SOCHO!

Q. (a) The variation of image distance (v) with object distance (u) for a ~~convex lens~~ is given in the following observation table. Analyse it and answer the questions that follow: (2024)

(on concave mirror)



S. No.	Object distance (u) cm	Image distance (v) cm
1	- 150	- 30
2	- 75	- + 37.5
3	- 50	- 50
4	- 37.5	- 75
5	- 30	- 150
6	- 15	+ + 37.5

$f = 25 \text{ cm}$ ✓

- Without calculation, find the focal length of the ~~convex lens~~. Justify your answer.
- Which observation is not correct? Why? Draw ray diagram to find the position of the image formed for this position of the object.
- Find the approximate value of magnification for $u = - 30 \text{ cm}$.

OR

- Define principal axis of a lens. Draw a ray diagram to show what happens when a ray of light parallel to the principal axis of a concave lens passes through it.
- The focal length of a concave lens is 20 cm . At what distance from the lens should a 5 cm tall object be placed so that its image is formed at a distance of 15 cm from the lens? Also calculate the size of the image formed.

**Kal Samjhenge
Refraction!**